

Elevating Haptics: An Accessible and Contactless Elevator Concept with Tactile Mid-Air Controls

Tanay Singhal
University of Waterloo
Waterloo, Ontario, Canada
tanay.singhal@uwaterloo.ca

Mahika Phutane
Cornell University
Ithaca, New York, United States
mahika@cs.cornell.edu

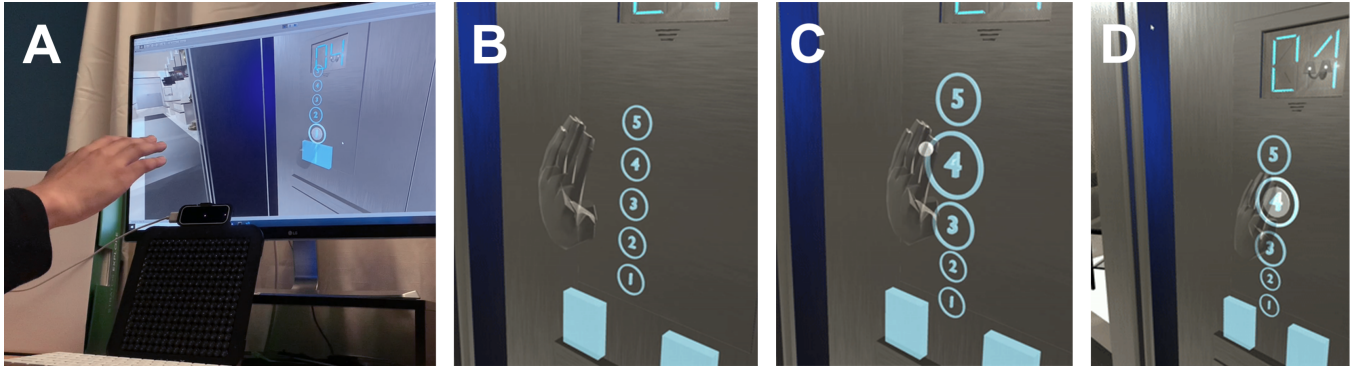


Figure 1: A: Demo setup of the contactless elevator simulation with mid-air haptics. The UltraHaptics device (black grid at the bottom) generates mid-air haptic feedback and a Leap Motion Controller above it tracks the user's hand. B: A screenshot of the simulation, prior to any interaction. C: As the hand approaches a button, the button magnifies and the user feels Braille information (visualized with the white dot), which is rendered Point-By-Point in mid-air. D: As the user pushes the button, they feel haptic sensations on their hand (visualized with the white circle).

ABSTRACT

Mid-air interfaces enable rich 3D interactions and are inherently hygienic, but the technology is not yet ready for real-world public interfaces such as elevators, ATMs, and kiosks – one barrier to their adoption is the limited work on the accessibility of mid-air systems. To this end, we designed an interactive simulation of a contactless elevator simulation with mid-air touch feedback and comprehensive accessibility considerations. Despite being fully contactless, the controls are tactile and closely resemble the mental model of ordinary elevator buttons. Moreover, our demo features 1) mid-air haptics for conveying Braille information, 2) responsive button magnification to assist people with low vision, 3) intuitive gestures for opening or closing doors, and 4) audio feedback. Because many of the ideas and interactions in this demo can apply to other mid-air public interfaces, we hope to inspire more work on improving the accessibility of mid-air interfaces.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility technologies**; **Haptic devices**; **Gestural input**.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8095-9/21/05.

<https://doi.org/10.1145/3411763.3451574>

KEYWORDS

mid-air haptics, public interface, accessibility, elevators

ACM Reference Format:

Tanay Singhal and Mahika Phutane. 2021. Elevating Haptics: An Accessible and Contactless Elevator Concept with Tactile Mid-Air Controls. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3411763.3451574>

1 INTRODUCTION

Today's public interfaces such as elevators, ATMs, and kiosks are touched by thousands of people per day in densely populated areas. We focus on elevators because they are essential to the lives of many – a single elevator can carry thousands of people per year. Even though their danger earned public attention during the COVID-19 pandemic [15], they have always been alarming sources of contamination [1]. Mid-air interfaces provide a hygienic alternative to traditional touch screens or buttons: they enable rich and intuitive interactions in a 3D space without having to touch a surface.

However, mid-air interfaces are far from public adoption due to sparse research on making them accessible, especially to people with visual impairments. Public interfaces are required to be accessible by law in many countries, including the United States where all public accommodations must comply with the Americans with Disabilities Act (ADA) regulations. For people who rely on their sense of touch, the lack of contact in mid-air interfaces is precisely what makes them less accessible.

Recent advances in mid-air haptics [3, 10]—technology that produces ultrasonic touch sensations in mid-air—present a promising solution for designing accessibility in mid-air interfaces. Inspired by Paneva & Long’s research [14] on conveying Braille information with mid-air haptics, we designed a holistic and accessible mid-air interaction prototype with a real-world application: elevators.

Using Unity3D, UltraHaptics STRATOS Explore, and the Leap Motion, we created a simulation (see Figure 1) of an elevator with only mid-air interactions – all of which are tactile. The demo features 1) mid-air touch feedback on contact with a button, 2) mid-air haptics to transmit Braille information, 3) responsive button magnification to assist people with low vision, 4) intuitive gestures for opening or closing doors, and 5) audio feedback. To our best knowledge, this is the first mid-air interactive experience with accessibility considerations in a real-world application.

2 BACKGROUND

2.1 Mid-Air Haptics

In 2013, Carter et al. introduced *UltraHaptics* [3], a system designed to provide discrete points of tactile feedback in mid-air through a grid of ultrasonic transducers. Kappus & Long expanded the technology in 2018 through *spatiotemporal modulation* [10], a technique for rendering volumetric 3D touch sensations in mid-air (as opposed to discrete points) via rapid translations of the ultrasonic focus point. The device we use in our demo—the UltraHaptics STRATOS™ Explore—supports *spatiotemporal modulation*.

Mid-air haptics present an immense potential for mid-air interfaces and many applications for the technology have been proposed. These include improving automotive user interfaces [8, 18], designing interactive digital signage [6], and devising a holographic retail shopping experience [11].

However, the accessibility of mid-air interfaces is largely unexplored. A key exception is the recent work by Paneva & Long, who studied three different methods for using mid-air haptics to convey Braille information to blind participants and found that generating the Braille dot information Point-by-Point (one dot at a time) had the highest average recognition accuracy (88%) [14] compared to Row-by-Row (one row of dots at a time) and Constant (all points at once). We draw from their design and results to transmit Braille information (see subsection 4.2).

Although Paneva & Long studied Braille information in isolation, their blind participants “had an overwhelmingly positive reaction to it and provided a list of scenarios and concrete tasks in their everyday life that could be facilitated by such device” such as ATMs, navigation maps, restrooms, and elevators. This is precisely the design space that our demo explores.

2.2 Designing for Accessibility in Public Interfaces

Accessibility is mediated by Assistive Technology (AT): technologies that are “used to increase, maintain or improve functional capabilities of individuals with disabilities.” [5, 19]. However, designing for accessibility in public spaces involves going beyond the *functional* requirements and also considering psycho-social factors. Shinohara & Wobbrock found that AT users preferred to avoid drawing special attention and wished to be viewed as professional

and technically savvy [19]. Bachofer reported that fear of stigmatization was a factor that dissuaded young adults with low vision from using optical device aids in public [2]. Our design takeaway is that accessibility should be built into a system’s main interaction flow, as opposed to special procedures that might stigmatize users and draw unwanted attention.

Multisensory feedback is instrumental when designing for accessibility because different populations rely on different senses. Díaz et al. found that the synergy of visual, auditory, and haptic stimuli led to improvements in accessibility tasks [7]. In their guidelines for designing accessible kiosks, Lazar et al. caution against relying on a single sensory characteristic when providing instructions or status indicators [12].

Past work has focused on the accessibility of touch screen interfaces in public settings. In 2002, Vanderheiden [20] introduced a touch screen system that is accessible to people with visual impairments. Since then, designs for self-serve machines (e.g., ATMs, kiosks) have been proposed with ATs such as haptic prompts [4] and gesture inputs [17]. To enable touch screen access for those with visual impairments without making hardware changes, Kane et al. [9] provided a set of gestures for exploring large displays such as public interactive maps. We use findings from these works to design for accessibility in our mid-air elevator simulation.

3 DEMO IMPLEMENTATION: HARDWARE AND DEVELOPER TOOLS

We used the UltraHaptics STRATOS™ Explore device to produce mid-air tactile sensations. The device consists of a 16x16 grid of ultrasonic transducers that can produce touch feedback in mid-air. A Leap Motion attached to the top of the device tracks a user’s hands in 3D space and is used to target tactile sensations. See Figure 1 for an image of the setup. We developed the demo using Unity 2019.1.14f1 and the C# programming language, along with the UltraHaptics SDK and Leap Motion SDK. Unity’s graphic display simulates being inside an elevator and interacting with holographic buttons. We also use *spatiotemporal modulation* [10] to render volumetric 3D touch sensations in mid-air. The complete open source code with setup instructions is available here: <https://github.com/TanaySinghal/Contactless-Elevator>.

4 INTERACTION DESIGNS

This section describes our interaction designs to create fully contactless (mid-air), tactile, and accessible elevator controls. We came up with these interaction designs through a review of accessibility research and informal discussions with people who have low vision or low mobility. Importantly, all accessibility features—such as mid-air Braille simulation and magnification—are ingrained into the main interaction flow that all users will experience. Doing so avoids special accommodations that might draw unwanted attention (see Figure 2).

4.1 Tactile Feedback on Button Press

Even though the buttons are virtual and in mid-air, they fit the mental model of a traditional button push interaction: as the user presses our mid-air elevator button, they will receive mid-air touch feedback on their hand. At the current state of mid-air haptics,

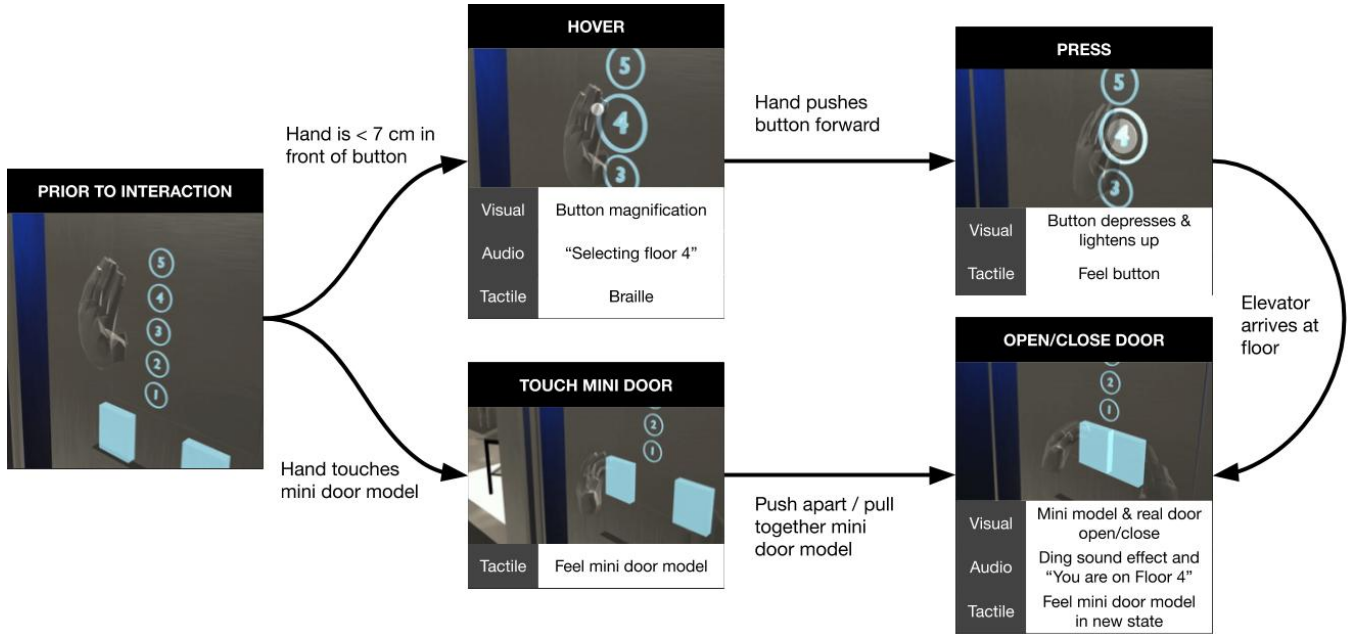


Figure 2: A diagram of the full interaction flow of the mid-air elevator controls. Accessibility features such as mid-air Braille simulation, magnification, and audio are built into the main interaction flow that are experienced by everyone.

feedback intensity is inversely proportional to surface area. For this reason, we concentrate the touch sensations along the contours of the buttons. According to Pei & Bensmaia, human perception of tactile motion is primarily predicted by a shape’s edges and terminators (corners) [16].

4.2 Mid-Air Haptics for Conveying Braille

Braille labels are mandatory for elevator buttons under ADA law so that people with visual impairments can independently navigate an elevator. Our elevator simulation uses mid-air haptics to convey Braille information, which is activated when a person’s hand is within 7 cm from a button. The dot information is generated in mid-air using Paneva & Long’s Point-by-Point technique (one dot at a time) [14]. Unlike Paneva & Long, we transmit braille information in the more conventional 6-dot Braille cell format instead of the 4-dot format. Like Paneva & Long, adjacent dots are 3 cm apart and individual dots are rendered for 0.2s, with a 0.3s pause between subsequent dots and a 0.5s pause at the end of a character. If there are N digits on a button and digit i takes d_i dots, the total reading time of a number is $0.5(N - 1) + \sum_{i=1}^N [0.2d_i + 0.3(d_i - 1)]$. For example, the total reading time for the number 3 ($N = 1, d_1 = 6$) is 2.7s and digit 27 ($N = 2, d_1 = 6, d_2 = 8$) is 4.2s.

4.3 Magnification on Hover

In addition to conveying Braille information, a person’s hand being within 7cm away from the device causes the closest elevator button to magnify. Besides serving an aesthetic purpose, the increased interactive surface area from magnification improves button-press

accuracy¹ [13] and character legibility, which is especially helpful for people with low vision. Whether for functional or aesthetic purposes, magnification is experienced by everybody (see subsection 2.2).

4.4 Audio Feedback

Redundant multisensory feedback is recommended when designing for accessibility because different populations rely on different senses [7, 12] (see subsection 2.2). In our demo, we generate sound effects and verbal descriptions for interactions and states. For instance, when a hand hovers near the button for floor 2, a pre-recorded voice says, “selecting floor 2”. When the elevator arrives at floor 2, a *ding* sound is heard and the voice informs, “you are on floor 2.” The audio feedback also describes the door’s status (“doors opening” and “doors closing”).

4.5 Intuitive Gestures: Opening and Closing Doors

Unlike their physical counterparts, virtual interfaces are flexible and dynamic. For example, touch screen keyboards on smartphones can adapt to a user’s need – they transform into an emoji keyboard, change languages, or disappear when not needed. We demonstrate the benefits of the virtual nature of mid-air interfaces: instead of scanning for an “open” or “close” button (and possibly mistaking them), we can manipulate a tangible holographic mini-model of the doors. The mini model’s doors can be pulled apart or pushed together to intuitively open or shut the real elevator’s door.

¹ Although our demo only includes five elevator buttons, button magnification would be more beneficial in elevators with many small and crowded buttons such as in a skyscraper.

5 PRELIMINARY FEEDBACK AND FUTURE WORK

We have not yet conducted a formal evaluation of the concept due to COVID-19 restrictions. Also, even though we designed this demo to be as accessible as possible, we cannot claim that our solution works for the complete spectrum of human abilities. Nevertheless, our demo has received positive preliminary feedback from our personal and professional network, especially from people with low vision and low mobility:

“The accessibility features for visual impairments seem really good, in particular. The magnification will help everyone, but especially people who are partially sighted. I’m also really curious to try out the haptic Braille.”
“I think it’s brilliant and, as a handicapped person, on behalf of the handicapped community, thank you”
“Yuck, I don’t know how I’m going to use standard elevators now. I don’t want to touch the Braille numbering.”

There remains a large and exciting opportunity for research on the design and user experience of mid-air interfaces in real-world contexts, especially regarding accessibility and usability. Future research should focus improving the speed of transmitting Braille characters and reduce the distances between adjacent dots. The 3D interaction space of mid-air interfaces is also ripe for design explorations. For instance, we can imagine elevator controls that move directly to passengers to avoid crowding in a corner and encourage social distancing.

6 CONCLUSION

We presented a design for an accessible mid-air elevator panel. Through the synergy of visual, audio, and tactile feedback, the elevator panel is designed to be fully contactless (i.e., hygienic) and accessible to people with little to no vision. Finally, we capitalize on the 3D interaction space of mid-air technology to create novel interactions such as opening and closing elevator doors. We encourage future researchers and designers to iterate on these ideas and improve the accessibility of mid-air interfaces.

ACKNOWLEDGMENTS

We acknowledge Oliver Schneider and Shiri Azenkot for their mentorship and feedback. We also thank Chris Barclays – as a person with low vision, Chris provided invaluable insight and advice throughout our design process. This project received support from Canada Foundation for Innovation’s (CFI) John R. Evan’s Leaders Fund.

REFERENCES

- [1] AK Al-Ghamdi, SMA Abdelmalek AM Ashshi, H Faidah, H Shukri, and AA Jiman-Fatani. 2011. Bacterial contamination of computer keyboards and mice, elevator buttons and shopping carts. *African Journal of Microbiology Research* 5, 23 (2011), 3998–4003. <https://doi.org/10.5897/AJMR11.770>
- [2] Cynthia Susan Bachofer. 2013. *Long-term optical device use by young adults with low vision*. Ph.D. Dissertation. Vanderbilt University, Nashville, TN, USA. <https://eric.ed.gov/?id=ED554976>
- [3] Tom Carter, Sue Ann Seah, Benjamin Long, Bruce Drinkwater, and Sriram Subramanian. 2013. UltraHaptics: Multi-Point Mid-Air Haptic Feedback for Touch Surfaces. In *Proceedings of the 26th Annual ACM Symposium on User Interface Software and Technology* (St. Andrews, Scotland, United Kingdom) (UIST ’13). Association for Computing Machinery, New York, NY, USA, 505–514. <https://doi.org/10.1145/2501988.2502018>
- [4] Brendan Cassidy, Gilbert Cockton, and Lynne Coventry. 2013. A Haptic ATM Interface to Assist Visually Impaired Users. In *Proceedings of the 15th International ACM SIGACCESS Conference on Computers and Accessibility* (Bellevue, Washington) (ASSETS ’13). Association for Computing Machinery, New York, NY, USA, Article 1, 8 pages. <https://doi.org/10.1145/2513383.2513433>
- [5] Albert Cook and Susan M. Hussey. 2002. *Assistive Technologies: Principles and Practice, 2nd edition*. Mosby, St. Louis, MO, USA.
- [6] Loic Corenthy, Marcello Giordano, Richard Hayden, Daniel Griffiths, Craig Jeffrey, Hannah Limerick, Orestis Georgiou, Tom Carter, Jörg Müller, and Sriram Subramanian. 2018. Touchless Tactile Displays for Digital Signage: Mid-Air Haptics Meets Large Screens. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI EA ’18). Association for Computing Machinery, New York, NY, USA, 1–4. <https://doi.org/10.1145/3170427.3186533>
- [7] Iñaki Diaz, Josune Hernantes, Ignacio Mansa, Alberto Lozano, Diego Borro, Jorge Juan Gil, and Emilio Sánchez. 2006. Influence of Multisensory Feedback on Haptic Accessibility Tasks. *Virtual Real.* 10, 1 (May 2006), 31–40. <https://doi.org/10.1007/s10055-006-0028-4>
- [8] Kyle Harrington, David R. Large, Gary Burnett, and Orestis Georgiou. 2018. Exploring the Use of Mid-Air Ultrasonic Feedback to Enhance Automotive User Interfaces. In *Proceedings of the 10th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Toronto, ON, Canada) (AutomotiveUI ’18). Association for Computing Machinery, New York, NY, USA, 11–20. <https://doi.org/10.1145/3239060.3239089>
- [9] Shaun K. Kane, Meredith Ringel Morris, Annuska Z. Perkins, Daniel Wigdor, Richard E. Ladner, and Jacob O. Wobbrock. 2011. Access Overlays: Improving Non-Visual Access to Large Touch Screens for Blind Users. In *Proceedings of the 24th Annual ACM Symposium on User Interface Software and Technology* (Santa Barbara, California, USA) (UIST ’11). Association for Computing Machinery, New York, NY, USA, 273–282. <https://doi.org/10.1145/2047196.2047232>
- [10] Brian Kappus and Ben Long. 2018. Spatiotemporal modulation for mid-air haptic feedback from an ultrasonic phased array. *The Journal of the Acoustical Society of America* 143, 3 (2018), 1836–1836. <https://doi.org/10.1121/1.5036027>
- [11] Jin Ryoung Kim, Stephanie Chan, Xiangchao Huang, Kenneth Ng, Limin Paul Fu, and Chen Zhao. 2019. Demonstration of Refinity: An Interactive Holographic Signage for New Retail Shopping Experience. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI EA ’19). Association for Computing Machinery, New York, NY, USA, 1–4. <https://doi.org/10.1145/3290607.3313269>
- [12] Jonathan Lazar, J. Bern Jordan, and Gregg Vanderheiden. 2019. Toward Unified Guidelines for Kiosk Accessibility. *Interactions* 26, 4 (June 2019), 74–77. <https://doi.org/10.1145/3337779>
- [13] Michael J. McGuffin and Ravin Balakrishnan. 2005. Fitts’ Law and Expanding Targets: Experimental Studies and Designs for User Interfaces. *ACM Trans. Comput.-Hum. Interact.* 12, 4 (Dec. 2005), 388–422. <https://doi.org/10.1145/1121112.1121115>
- [14] Viktorija Paneva, Sofia Seinfeld, Michael Kraicz, and Jörg Müller. 2020. HaptiRead: Reading Braille as Mid-Air Haptic Information. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference* (Eindhoven, Netherlands) (DIS ’20). Association for Computing Machinery, New York, NY, USA, 13–20. <https://doi.org/10.1145/3357236.3395515>
- [15] Tara Parker-Pope. 2020. *Can I Get Coronavirus From Riding an Elevator?* The New York Times. <https://www.nytimes.com/2020/05/13/well/live/can-i-get-coronavirus-from-riding-an-elevator.html>
- [16] Yu-Cheng Pei and Sliman J. Bensmaia. 2014. The neural basis of tactile motion perception. *Journal of Neurophysiology* 112, 12 (2014), 3023–3032. <https://doi.org/10.1152/jn.00391.2014>
- [17] Frode Eika Sandnes, Tek Beng Tan, Anders Johansen, Edvin Sulic, Eirik Vesterhus, and Eirik Rud Iversen. 2012. Making touch-based kiosks accessible to blind users through simple gestures. *Universal Access in the Information Society* 11, 4 (2012), 421–431.
- [18] Gözel Shakeri, John H. Williamson, and Stephen Brewster. 2018. May the Force Be with You: Ultrasound Haptic Feedback for Mid-Air Gesture Interaction in Cars. In *Proceedings of the 10th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Toronto, ON, Canada) (AutomotiveUI ’18). Association for Computing Machinery, New York, NY, USA, 1–10. <https://doi.org/10.1145/3239060.3239081>
- [19] Kristen Shinohara and Jacob O. Wobbrock. 2016. Self-Conscious or Self-Confident? A Diary Study Conceptualizing the Social Accessibility of Assistive Technology. *ACM Trans. Access. Comput.* 8, 2, Article 5 (Jan. 2016), 31 pages. <https://doi.org/10.1145/2827857>
- [20] Gregg C Vanderheiden. 2002. Touch screen for the vision-impaired. US Patent 6,384,743.